Brain tumor detection using image process in Deep Learning

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Referred:

A Deep Learning Model Based on Concatenation Approach for the Diagnosis of Brain Tumor

Multimodal data fusion of adult and pediatric brain tumors with deep learning

A Deep Learning-Based Framework for Automatic Brain Tumors Classification Using Transfer Learning

An Evolutionary Deep Learning Method Based on Improved Heap-Based Optimization

Detection of tumors on brain MRI images using the hybrid convolutional neural network architecture

Abstract:

The disease of cancer different data modalities may reveal different disease characteristics, and the integration of imaging with genomic data has the potential to unravel additional information then when using these data sources in isolation. The presented work is a pioneer study in the domain of brain tumor classification using transfer learning and deep CNN architectures. We applied transfer learning techniques using natural images of ImageNet dataset (source task) and classified the brain tumor type from glioma, meningioma, and pituitary using Figshare dataset (target task). Consequently, the ensemble method based on concatenation of dense block by using DensNet201 pre-trained model outperformed as compared to the current research methods for brain tumor classification problem. The proposed method produced 99.51% testing accuracy on testing samples and achieved the highest performance in detection of brain tumorThen, these features were concatenated and passed to softmax classifier to classify the brain tumor. Secondly, the features from different Inception modules were extracted from pre-trained Inceptionv3 model and concatenated and then, passed to the softmax for the classification of brain tumors. Both scenarios were evaluated with the publicly available three-class brain tumor dataset.

Using deep learning method:

Deep Belief Networks

- **Data Collection:**
  - Gather a dataset of brain MRI images. You can use publicly available datasets like the Brain Tumor Segmentation (BraTS) dataset.
- **Preprocessing:**
  - Normalize the MRI images to ensure consistent input data.
  - Apply data augmentation techniques to increase the diversity of the training data and prevent overfitting.
- **Segmentation:**
  - Use image segmentation techniques to isolate the tumor region from the rest of the brain. Techniques like hybrid active contour models can be effective[1].
- **Feature Extraction:**
  - Extract relevant features from the segmented tumor regions. This can include texture, shape, and intensity features.
- **Deep Belief Network (DBN) Model:**
  - Build a DBN model composed of multiple layers of Restricted Boltzmann Machines (RBMs).
  - Train the DBN on the extracted features to learn the underlying patterns associated with different types of brain tumors.
- **Classification:**
  - Use the trained DBN to classify the MRI images into different categories (e.g., benign or malignant tumors).
  - Evaluate the model's performance using metrics like accuracy, sensitivity, and specificity.
- **Validation and Testing:**
  - Validate the model using a separate validation dataset.
  - Test the model on unseen data to assess its generalization capability.
- **Deployment:**
  - Develop a user-friendly interface where medical professionals can upload MRI images and receive predictions about the presence and type of brain tumor.

  - Consider deploying the model as a web application for easy access.

Conclusion:

- A trained DBN model capable of accurately detecting and classifying brain tumors from MRI images.
- A web application that provides an interface for medical professionals to use the model for diagnostic purposes.